

SNAPFRAME · DESIGN DECISIONS REGISTER

SnapFrame design decisions register

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DOCUMENT CONTROL

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Project TRL	TRL 3 Proof of concept		
Repository	github.com/BoujeeEnjinia1701/snapframe		
Source	docs/06-design-decisions.md @ uncommitted		

REVISION HISTORY

Version	Date	Author	Change
0.1	2026-10-01	Amish Chadha	Register opened with the build plan; open items from SNF-DDR-001 to SNF-DDR-003 and the review note; budget treated as a value-engineering target

SnapFrame design decisions register

Every design decision still to be made, and every decision made, in one place. Each decision is argued in full in its decision record under `docs/decisions/` ; this register is the index Amish works from. The build plan (`docs/05-build-plan.md`) describes the design as it stands and does not list open decisions.

■ Open decisions

#	Decision needed	Options	Recommendation	Affects in the build	Source
1	Accept the design for construction: button 40 mm and hitch pin 15 mm from the tube end, finger recesses and mouth chamfers, through-bolted cable bolts with rings in place of printed tabs, cables clipped to the anchor eyes, anchor eyes down on the plate, two anchor slots, frames slid onto the tubes before anchoring, one drilling rule for every tube	Accept C1 to C9; change any of them	Accept all nine	Every node and tube, the cables and the erection order (build plan sections 3 and 4)	SNF-DDR-003, C1 to C9
2	Guiding the ridge tubes at 2.6 m when a frame slides onto them (R4, no tools)	(a) folding step in the kit, about 1.5 kg; (b) guide from the ground with a spare tube; (c) build the roof low and lift it onto the posts	(b), timed in the TRL 4 user trial; the first prototype uses a step	Steps 5 to 7 of the build plan	SNF-DDR-003, A1
3	Open front gable (R7): two tarpaulins close the roof, side walls and one gable only. Includes how the side wall tarpaulin passes the cables at the corner anchors	Cutting and folding plan; a third tarpaulin; accept one open gable as the doorway	None yet; for the corner cables, settle them in the same tarpaulin plan	Step 13 (tarpaulins)	SNF-DDR-001, O3; SNF-DDR-003, A2
4	1 in posts (R6): the 3/4 in posts reach a factor of 1.46 at 20 m/s; 1 in posts would give 20.3 m/s by member check, about \$30 more	Decide after the frame analysis named in SNF-DDR-002 D8	Run the frame analysis first	Posts, foot nodes and eave node post sockets	SNF-DDR-002, O8
5	Packages and carried mass (R5, R11): tube bundle 28.6 kg against 25 kg; complete kit with tarpaulins 52.5 kg against 50 kg	Two tube bundles (16.5 and 12.1 kg); relax R5 to 30 kg for a two-person carry; and a choice for R11	None yet	Packing only; not the prototype build	SNF-DDR-002, O5; SNF-DDR-003
6	Node polymer (R9): printed sockets at 70 °C carrying 32 MPa of pin bearing with the factor of 2	ASA, PETG, nylon blends; later cast aluminium	None yet	Printing of every node (section 3 of the build plan assumes ASA class)	SNF-DDR-002, O6
7	Anchor target (R8): 1.0 kN target against 1.28 kN calculated demand at a rear corner foot	Raise the target; longer anchors at the rear corners; more pretension in the guys	None yet	Anchor choice	SNF-DDR-002, O7

#	Decision needed	Options	Recommendation	Affects in the build	Source
8	Tube offcuts: 39 % of the tube bought is offcut	Accept; shorten posts to 1.50 m; buy 6 m metric stock	Decide once the first region's tube source is known	Tube cutting list	SNF-DDR-001, O1
9	First co-design partner and region	Not yet listed	None yet	None in the prototype build	SNF-DDR-001, O2
10	Appearance model choices from the render session: tube split at the detail node as a render device; hero without guy lines; foot plate grip ribs and size mark as appearance only; brace cable start at the detail node	Accept each, or change the renders	Accept the first three; the fourth is overtaken by the cable bolt (SNF-DDR-003, C3) and is withdrawn when the renders are redone	Photoreal renders only	docs/ REVIEW.md , 2026-09-26 render session, items 1 to 4

■ To confirm when parts are bought

#	What to confirm	Why it matters	Source
1	Snap buttons for 3/4 in and 1 in EMT: the spring reaches no more than 45 mm deeper than the button, and the button stands at least 3.5 mm above the tube surface	The spring must clear the hitch pin 15 mm from the tube end, and the button must stand proud in its finger recess	SNF-DDR-003, C1 and C2
2	Snap hooks: the gate opens at least 14 mm, to pass the 12 mm wire of the anchor eye, and they fit the 6 mm ring	Every cable end clips to an anchor eye or a ring	SNF-DDR-003, C3 and C4
3	Welded 6 mm ring of 32 mm bore: working load at least 2 kN; hand tensioner for 4 mm wire: working load at least 1.5 kN	The highest cable tension is 1.21 kN at 20 m/s; the tensioner rating has been unknown since TRL 3	SNF-CAL-001, section 6
4	Screw anchor: eye wire about 12 mm and eye about 56 mm across, so it straddles the 18 mm slot; holding capacity in the soils of the first site	The eye clamps the foot plate; R8 needs 1.0 kN per anchor	SNF-DDR-003, C5; R8
5	EMT yield strength and dimensions from the supplier	The wind rating assumes 275 MPa yield	SNF-CAL-001, A3
6	Print farm bed at least 230 mm, and the polymer it can print	The eave and middle ridge nodes are 220 mm across	SNF-DDR-003; open decision 6

■ Value engineering

Value-engineering target: USD 445 (a hypothetical control target, not a limit). Estimated cost of the constructable design: USD 469.14 for the size M frame kit (USD 24.14 over the target). Tarpaulins are agency stock and costed separately (USD 50.00). Main cost drivers and savings worth trying:

- The largest lines are the tubes (USD 202.00 for 18 sticks, 39 % of the tube bought ends up as offcut), the 15 printed nodes (USD 119.74, of which the six foot nodes are USD 60.18), the brace cables (USD 42.00), the anchors (USD 32.00), the buttons, pins and caps (USD 24.00) and the cable bolt sets (USD 23.40).

- Making the design constructable added the cable bolt sets (USD 23.40) and a second snap hook on every cable (USD 7.00), and saved USD 4.56 of print; the frame kit rose from USD 443.30 to USD 469.14.
- Savings worth trying: bulk or 6 m metric tube, which also cuts the offcut (open decision 8); a lighter foot plate, since the foot node is the most expensive print; lower infill on the low-stress ridge nodes once the polymer is chosen (open decision 6); and stainless hardware bought by the hundred for the cable bolt sets.

■ Decisions made

Date	Decision	Decided by	Record
2026-09-25	TRL 2 review items D1 to D7: budget covers the frame kit, tarpaulins from agency stock; 1 in EMT rafters; print nodes, design for casting; spring buttons plus hitch pins at the 12 tension joints; gable with cable bracing; sizes S, M and L with M as reference; snow and cyclone out of scope	Amish: "proceed with all of your recommendations across all batches. Make sure we don't proceed to TRL 4 on any of them."	SNF-DDR-001
2026-09-25	D8: 1 in EMT ridge tubes, frame analysis before any decision on posts. D9: one four-socket eave node at all six positions, the corner socket past the gable left blank and capped	Amish: "i accept all your recommendations, go with them across all repos."	SNF-DDR-002
2026-09-26	Budget set to \$445 to cover the priced frame kit (O4 closed)	Amish: "i approve all the budget items."	SNF-DDR-002 v0.2
2026-09-30	Prototype build plan in the approved format; outstanding decisions kept out of the build plan and in this register	Amish: "this is the correct build plan ... this is a good quality document format. Extend this across all the other repos"; "don't log outstanding decisions in this build plan - that is not the place for it"	<code>.kit/STANDARDS.md</code> section 18
2026-09-30	Fix the design where it cannot be built as drawn, keeping what it does	Amish: "If you are realising that the design cannot be built as per concept - fix the design assumptions to match and be physically feasible as you draw the illustrations."	SNF-DDR-003 (the changes themselves are open decision 1)
2026-10-01	<code>budget_usd</code> is a value-engineering target, not a limit; cost reported as over or under it	Amish: "the budgets are a hypothetical control target to ensure we are thinking along a value engineering lens"	<code>.kit/STANDARDS.md</code> section 18